Supplementary Material for: Steering the Flow: Quality-Aware Personalized Prior Construction for Generative Collaborative Filtering

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A Theoretical Analysis Details

A.1 Proof of Theorem 4.1

We formalize the core requirement for the generative trajectory through the following assumption.

Assumption 1 (γ-Contraction). *The trained flow model f_θ satisfies a contraction property with respect to the target $\mathbf{x}_1$. There exists $\gamma \in (0,1)$ such that for all $t \in [0,1]$ and all states $\mathbf{x}$,*

$$\|f_\theta(\mathbf{x},\, t) - \mathbf{x}_1\| \;\le\; \gamma\, \|\mathbf{x} - \mathbf{x}_1\|. \tag{1}$$

Proof. We begin by formally expressing the discrete Euler integration update. At iteration $k \in \{0, 1, \ldots, N-2\}$ with timestep $t_k = k/N$, the state transition is defined precisely as

$$\mathbf{x}_{t_{k+1}} = \mathbf{x}_{t_k} + \frac{1}{N} \cdot \frac{\hat{\mathbf{x}}_1^{(k)} - \mathbf{x}_{t_k}}{1 - t_k} = \frac{N-k-1}{N-k}\, \mathbf{x}_{t_k} + \frac{1}{N-k}\, \hat{\mathbf{x}}_1^{(k)}, \tag{2}$$

where $\hat{\mathbf{x}}_1^{(k)} = f_\theta(\mathbf{x}_{t_k}, t_k)$ denotes the model prediction at the current state. We define the absolute error vector as $\mathbf{d}_k = \mathbf{x}_{t_k} - \mathbf{x}_1$. Subtracting the target $\mathbf{x}_1$ from both sides of Eq. (2) yields the rigorous error transition

$$\mathbf{d}_{k+1} = \frac{N-k-1}{N-k}\, \mathbf{d}_k + \frac{1}{N-k}\big(\hat{\mathbf{x}}_1^{(k)} - \mathbf{x}_1\big). \tag{3}$$

Applying the structural inequality from Assumption 1, we obtain $\|\hat{\mathbf{x}}_1^{(k)} - \mathbf{x}_1\| \le \gamma \|\mathbf{d}_k\|$. Substituting this directly into the norm of Eq. (3) produces the recursive bound

$$\|\mathbf{d}_{k+1}\| \;\le\; \frac{N-k-1+\gamma}{N-k}\, \|\mathbf{d}_k\|. \tag{4}$$

Iterating this inequality systematically from the initial state $k = 0$ to the penultimate state $k = N-2$ gives

$$\|\mathbf{d}_{N-1}\| \;\le\; \prod_{k=0}^{N-2} \frac{N-k-1+\gamma}{N-k}\, \|\mathbf{d}_0\|. \tag{5}$$

By applying the mathematical substitution $l = N-k-1$, we systematically rewrite the product terms. We continuously expand and bound the final integrated target distance through the following complete sequence:

$$\begin{aligned}
\big\|\hat{\mathbf{x}}_1^{(N)} - \mathbf{x}_1\big\| &\le \gamma\, \|\mathbf{d}_{N-1}\| \\
&\le \gamma \left(\prod_{l=1}^{N-1} \frac{l+\gamma}{l+1} \right) \|\mathbf{x}_0 - \mathbf{x}_1\| \\
&= \gamma \left(\frac{\Gamma(N+\gamma)}{\Gamma(1+\gamma)\, \Gamma(N+1)} \right) \|\mathbf{x}_0 - \mathbf{x}_1\| \\
&= \gamma \left(\frac{\Gamma(N+\gamma)}{\gamma\, \Gamma(\gamma)\, \Gamma(N+1)} \right) \|\mathbf{x}_0 - \mathbf{x}_1\| \\
&= \left(\frac{\Gamma(N+\gamma)}{\Gamma(\gamma)\, \Gamma(N+1)} \right) \|\mathbf{x}_0 - \mathbf{x}_1\| \\
&\le \left(\frac{N^{\gamma-1}}{\Gamma(\gamma)} \right) \|\mathbf{x}_0 - \mathbf{x}_1\| \\
&= \frac{N^{-(1-\gamma)}}{\Gamma(\gamma)}\, \|\mathbf{x}_0 - \mathbf{x}_1\|.
\end{aligned} \tag{6}$$

Squaring both sides and taking the expectation over $\mathbf{x}_0$ strictly establishes the bound in part (a).

To establish part (b), we rigorously evaluate the expected source-target distance $\Delta^2(\boldsymbol{\mu}_u)$. The personalized prior is formulated as $\boldsymbol{\mu}_u = \mathbf{f} + \alpha_u(\boldsymbol{\mu}_u^{\text{nb}} - \mathbf{f})$. Since the ground-truth interactions are strictly binary ($x_{1,i} \in \{0,1\}$), the mathematical identity $x_{1,i} = x_{1,i}^2$ perfectly holds. We continuously expand and reorganize the bias-variance formulation of $\Delta^2(\boldsymbol{\mu}_u)$ through the following rigorous derivations:

$$\begin{aligned}
\Delta^2(\boldsymbol{\mu}_u) &= \sum_{i=1}^{|\mathcal{I}|} \Big((\mu_{u,i} - x_{1,i})^2 + \mu_{u,i}(1 - \mu_{u,i}) \Big) \\
&= \sum_{i=1}^{|\mathcal{I}|} \Big(\mu_{u,i}^2 - 2\mu_{u,i} x_{1,i} + x_{1,i}^2 + \mu_{u,i} - \mu_{u,i}^2 \Big) \\
&= \sum_{i=1}^{|\mathcal{I}|} \Big(x_{1,i}^2 - \mu_{u,i}(2x_{1,i} - 1) \Big) \\
&= \sum_{i=1}^{|\mathcal{I}|} \Big(x_{1,i}^2 - \big(f_i + \alpha_u(\mu_{u,i}^{\text{nb}} - f_i)\big)(2x_{1,i} - 1) \Big) \\
&= \sum_{i=1}^{|\mathcal{I}|} \Big(x_{1,i}^2 - f_i(2x_{1,i} - 1) - \alpha_u(\mu_{u,i}^{\text{nb}} - f_i)(2x_{1,i} - 1) \Big) \\
&= \Delta^2(\mathbf{f}) - \sum_{i=1}^{|\mathcal{I}|} \alpha_u(\mu_{u,i}^{\text{nb}} - f_i)(2x_{1,i} - 1) \\
&= \Delta^2(\mathbf{f}) - 2\, \alpha_u \sum_{i=1}^{|\mathcal{I}|} (x_{1,i} - 0.5)(\mu_{u,i}^{\text{nb}} - f_i) \\
&= \Delta^2(\mathbf{f}) - 2\, \alpha_u \big\langle \mathbf{x}_1 - \tfrac{1}{2}\mathbf{1},\; \boldsymbol{\mu}_u^{\text{nb}} - \mathbf{f} \big\rangle.
\end{aligned} \tag{7}$$

This structurally establishes the exact source-target decomposition formalized in part (b). Part (c) mathematically follows by dividing both sides of the exact decomposition by the baseline distance $\Delta^2(\mathbf{f})$. This completes the proof. □

A.2 Empirical Verification of Theorem 4.1

We directly compute the exact theoretical quantities from Theorem 4.1 on the real-world datasets. For each user u, we extract the personalized prior $\boldsymbol{\mu}_u$ directly from the trained QualCF model. We subsequently calculate $\Delta^2(\boldsymbol{\mu}_u)$, $\Delta^2(\mathbf{f})$, the precise alignment gain G_u, and the ratio ρ_u.

We verify the continuous algebraic identity in Eq. (**??**) by computing both sides independently. For all users across all datasets, the absolute numerical difference is strictly below 10^{-4}. Table 1 comprehensively summarizes the overall population-level statistics.

Table 1: Empirical verification of Theorem 4.1. $\mathbb{E}[\alpha_u]$: mean confidence weight. $\mathbb{E}[G_u]$: mean neighbor alignment gain. $G_u > 0$: percentage of users satisfying the condition. $\mathbb{E}[\rho_u]$: mean error reduction ratio.

Dataset	$\mathbb{E}[\alpha_u]$	$\mathbb{E}[G_u]$	$G_u > 0$	$\mathbb{E}[\rho_u]$	Reduction
Amazon-Beauty	0.999	2.08	94.9%	0.644	31.1%
ML-1M	0.984	4.81	66.7%	0.801	6.7%
ML-20M	0.645	12.64	83.0%	0.743	14.6%

To understand how the personalized prior quality varies dynamically across different user interaction densities, we explicitly partition the users into three distinct groups based on their historical interaction counts. Table 2 reports the exact mathematical decomposition statistics per group.

Table 2: Theorem verification partitioned by user activity group. G_u^+ and G_u^- functionally denote the positive-item and negative-item contributions.

Dataset	Group	N	$\mathbb{E}[G_u^+]$	$\mathbb{E}[G_u^-]$	$G_u > 0$	$\mathbb{E}[\rho_u]$
Amazon-Beauty	Cold	3377	0.5	1.5	96.8%	0.585
	Normal	3259	0.6	1.7	98.4%	0.583
	Active	3918	1.2	0.8	90.4%	0.744
ML-1M	Cold	1948	2.0	24.4	99.7%	0.422
	Normal	2032	5.0	9.0	86.7%	0.748
	Active	2055	22.4	−47.1	15.7%	1.213
ML-20M	Cold	44425	1.5	24.2	100.0%	0.525
	Normal	45546	3.3	17.6	99.3%	0.682
	Active	46704	16.7	−24.4	51.0%	1.009

Cold-start users consistently exhibit the largest mathematical alignment gain and nearly universal satisfaction of the reduction condition. For cold and normal users, the dominant contribution securely originates from the negative-item suppression G_u^-. The expected theoretical error ratio strictly drops due to the highly sparse and distinctive collaborative patterns. The population-level expectations mathematically verify that the personalized prior successfully tightens the error bound overall.

B Training Algorithm of QualCF

Algorithm 1 details the training process of QualCF.

Algorithm 1 Training QualCF

Require: Training interactions $\bar{X}$, flow model f_θ, quality network g_ω, number of discrete steps N, top-K neighbors, update frequency E_{update}.

Compute global item frequency $\mathbf{f}$ from $\bar{X}$.
Build initial user similarity index and retrieve top-K neighbors.

repeat
  Sample a batch of users $\mathcal{B}$ and set $\mathbf{x}_1 \leftarrow \bar{X}[\mathcal{B}]$.
  for each user $u \in \mathcal{B}$ **do**
    Compute features $\{\boldsymbol{\phi}_k\}_{k=1}^K$.
    Predict scores $q_k = g_\omega(\boldsymbol{\phi}_k)$ and weights w_k.
    Compute prior mean $\boldsymbol{\mu}_u$.
  end for
  Sample $\mathbf{x}_0 \sim \text{Bernoulli}(\boldsymbol{\mu}_u)$.
  Sample time steps $t \sim \text{Uniform}\{0/N, 1/N, \ldots, N/N\}$.
  Compute interpolation $\mathbf{x}_t$.
  Predict $\hat{\mathbf{x}}_1 = f_\theta(\mathbf{x}_t, t)$ and compute $\mathcal{L}$.
  Update θ, ω via gradient descent on $\nabla_{\theta,\omega}\mathcal{L}$.
  if epoch mod $E_{\text{update}} = 0$ **then**
    Rebuild user similarity index.
  end if
until converged

Ensure: Optimized parameters θ, ω.

C Inference Algorithm of QualCF

Algorithm 2 details the inference process of QualCF.

Algorithm 2 Inference with QualCF

Require: Trained flow model f_θ, quality network g_ω, user interactions $\mathbf{x}_u$, number of steps N, starting step s, retrieval weight β.

Retrieve top-K neighbors and compute prior $\boldsymbol{\mu}_u^{\text{nb}}$.
Initialize $\mathbf{x}_t \leftarrow \mathbf{x}_u$.
for $i = s$ to $N - 2$ **do**
  Set $t \leftarrow i/N$.
  Predict $\hat{\mathbf{x}}_1 = f_\theta(\mathbf{x}_t, t) + \beta \cdot \boldsymbol{\mu}_u^{\text{nb}}$.
  Compute $v_t \leftarrow (\hat{\mathbf{x}}_1 - \mathbf{x}_t)/(1 - t)$.
  Update $\mathbf{x}_t$.
  Preserve observed interactions: $\mathbf{x}_t \leftarrow \mathbf{x}_t \vee \mathbf{x}_u$.
end for
Predict final scores $\hat{\mathbf{x}}_1 = f_\theta(\mathbf{x}_t, t_{N-1}) + \beta \cdot \boldsymbol{\mu}_u^{\text{nb}}$.

Ensure: Predicted interaction scores $\hat{\mathbf{x}}_1$.

D Extended Experimental Details

In this section, we provide supplementary details regarding our experimental setup to facilitate reproducibility and offer deeper insights into the model's behavior. Specifically, we elaborate on the dataset statistics, summarize the evaluated baseline models, detail the comprehensive training protocols, and present a rigorous hyperparameter sensitivity analysis of our proposed QualCF.

D.1 Datasets.

We conduct our experiments on three widely adopted public benchmarks: Amazon-Beauty[1], MovieLens-1M[2], and MovieLens-20M[3]. The detailed statistics of the preprocessed datasets used in our evaluation are summarized in Table 3.

Table 3: Statistics of the experimented datasets.

Dataset	# Users	# Items	# Interactions	Density
Amazon-Beauty	10,553	6,087	94,148	0.15%
MovieLens-1M	6,034	3,126	574,376	3.04%
MovieLens-20M	136,674	13,681	9,977,451	0.53%

D.2 Baselines.

We compare our proposed QualCF with twelve representative baseline models, which comprehensively span classic collaborative filtering, graph neural networks, and state-of-the-art generative approaches:

- **BPR** [7]: A classic matrix factorization model optimized with the Bayesian Personalized Ranking pairwise loss for implicit feedback.
- **LightGCN** [1]: A collaborative filtering model that simplifies graph convolution networks by omitting feature transformation and nonlinear activation, retaining only neighborhood aggregation.
- **SGL** [10]: A GNN-based model that leverages self-supervised contrastive learning with graph augmentations (e.g., node or edge dropout) to alleviate data sparsity.
- **NCL** [5]: A neighborhood-enriched contrastive learning framework that explicitly incorporates potential structural and semantic neighbors into graph collaborative filtering.
- **DGCL** [3]: A disentangled graph contrastive learning method that models complex and diverse user intents to learn robust representations for recommendation.
- **EASE** [8]: A linear, embarrassingly shallow autoencoder model equipped with a closed-form solution, recognized for its simplicity and highly competitive performance.
- **MultiVAE** [4]: A generative model based on Variational Autoencoders (VAEs) that employs a multinomial likelihood to model implicit user-item interactions.
- **DiffRec** [9]: A diffusion-based generative model that infers user preferences by modeling interaction probabilities through a forward noise-adding and reverse denoising process.
- **LDiffRec** [9]: An extension of DiffRec that applies the diffusion and denoising processes within a compressed continuous latent space for improved efficiency and scalability.
- **GiffCF** [11]: A graph-enhanced generative diffusion model that leverages graph structures to guide the generation process and capture higher-order collaborative signals.
- **CDiff4Rec** [2]: A conditional diffusion model designed for recommendation that generates target item distributions conditioned on user interaction histories.
- **FlowCF** [6]: A state-of-the-art generative baseline utilizing a discrete Flow Matching (FM) framework to model user-item interactions. As QualCF directly builds upon this discrete FM framework, we treat it as our primary baseline for comparison.

[1]https://jmcauley.ucsd.edu/data/amazon/
[2]https://grouplens.org/datasets/movielens/1m/
[3]https://grouplens.org/datasets/movielens/20m/

D.3 Implementation Details.

QualCF is trained with the Adam optimizer using a 1e-3 learning rate and 4096 batch size. We apply early stopping with a patience of 10 on NDCG@10. While core architectural parameters remain constant across all datasets, specific hyperparameters are tuned based on validation performance. The final configurations are summarized in Table 4.

Table 4: Hyperparameter settings of QualCF.

Hyperparameter	Amazon-Beauty	ML-1M	ML-20M
Flow steps T	9	9	9
Sampling steps S	2	2	2
Time emb. size	10	10	10
Optimizer	Adam	Adam	Adam
Learning rate	1e-3	1e-3	1e-3
Batch size	4096	4096	4096
Early stopping	10	10	10
Hidden dim d	300	250	500
Dropout	0.2	0.2	0.15
Neighbor K	22	24	26
Retrieval freq	10	7	7

D.4 Hyperparameter Sensitivity.

To evaluate the robustness of QualCF under various conditions, we conduct a hyperparameter sensitivity analysis on three critical parameters: the number of retrieved neighbors K, the number of ODE integration steps T, and the hidden dimension of the Quality-Net d_q. Figure 1 illustrates the relative performance fluctuations across the three datasets.

The model demonstrates remarkable stability concerning the ODE integration steps T, with relative performance variations tightly constrained within a narrow margin of -2.5% to $+0.5\%$. This confirms that the deterministic vector field learned by flow matching enables accurate trajectory integration even with coarse discretization. Regarding the neighborhood size K, the maximum performance drop remains strictly bounded between -6.0% and -8.0% even at extreme boundary values. This corroborates the efficacy of the quality-aware selector in dynamically filtering uninformative signals regardless of the initial retrieval pool size. Finally, variations in the hidden dimension d_q reveal that while severe under-parameterization causes performance drops of up to -8.5% on ML-20M and Amazon-Beauty, the metrics cluster densely near the 0.0% optimal reference line under reasonable dimensional bounds. These bounded fluctuations highlight the generalization capability and minimal tuning cost of the QualCF framework.

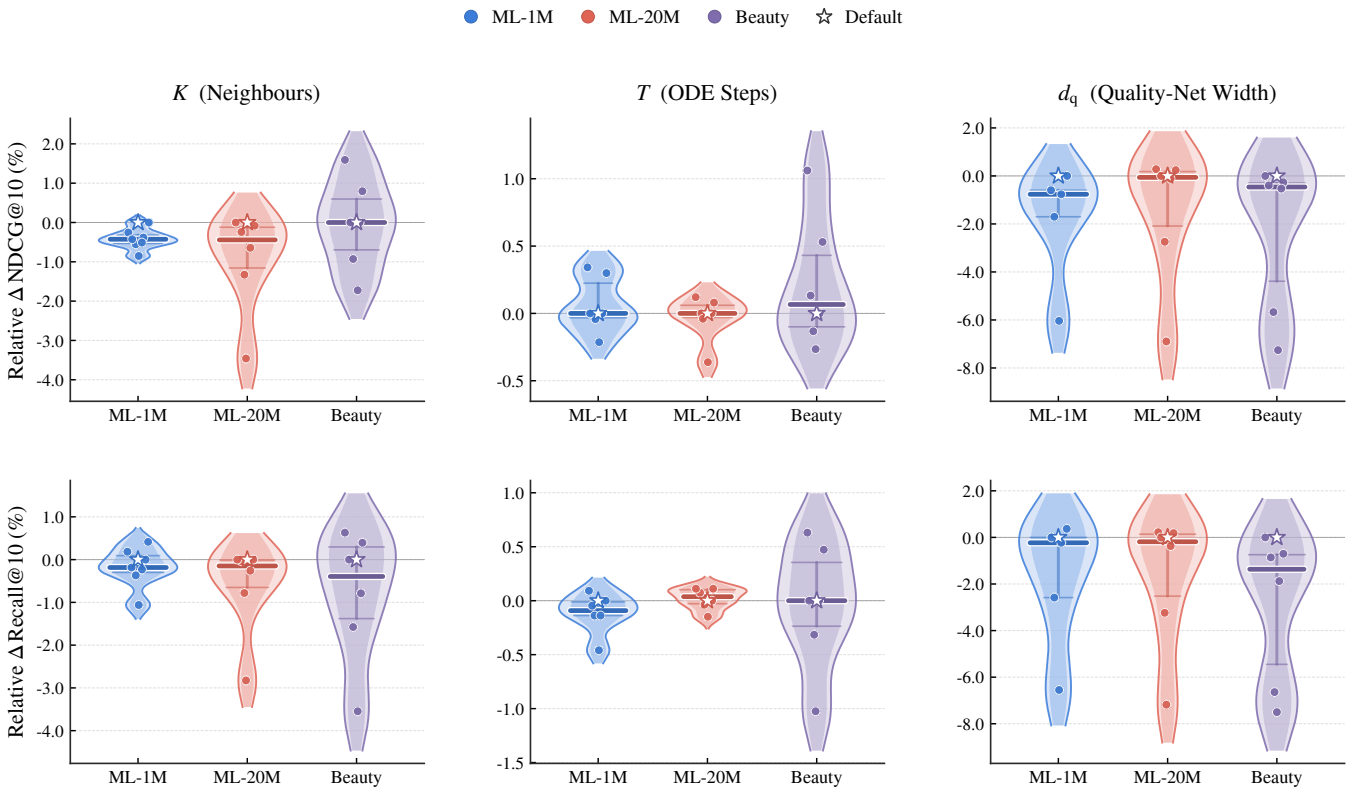

Figure 1: Hyperparameter sensitivity analysis of QualCF across three datasets. The symmetric violin plots illustrate the relative performance fluctuations in NDCG@10 and Recall@10 when varying the neighborhood size K, the number of ODE steps T, and the Quality-Net width d_q. The star marker denotes the default optimal configuration.

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